NITESH KUMAR

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SUMMARY

B.Tech Computer Science (AI & ML) student with hands-on experience building machine learning, NLP, computer vision, and backend applications using Python, PyTorch, Scikit-learn, HuggingFace Transformers, FastAPI, and OpenCV. Developed end-to-end ML pipelines from data preprocessing and model training to evaluation and deployment. Seeking an entry-level Software Engineer or ML Engineer role.

TECHNICAL SKILLS

Languages: Python, Java, SQL

ML / Data: PyTorch, Scikit-learn, Pandas, NumPy, HuggingFace Transformers, NLTK, OpenCV, Voice Recognition, Audio Processing

Web / Tools: HTML, CSS, JavaScript, FastAPI, Streamlit, Git & GitHub, VS Code

Core CS: Data Structures & Algorithms, OOP, DBMS basics

DevOps & Cloud: Kubernetes (K8s), CI/CD, MongoDB

Core Competencies: Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Supervised Learning & Classification, Model Training & Evaluation (Precision, Recall, F1-score, Confusion Matrix), Text Preprocessing & TF-IDF, Natural Language Processing, Computer Vision, Model Deployment, Version Control, Multi-modal Biometrics, Problem Solving

PROJECTS

SpamShield — End-to-End SMS Spam Detection Pipeline *Personal Project*

- Built an end-to-end SMS spam classification pipeline covering data scraping, NLP preprocessing, model training, and deployment.
- Cleaned and processed raw SMS data using Pandas and NumPy, applying NLP techniques (tokenization, stopword removal, TF-IDF) with NLTK for feature extraction.
- Trained and benchmarked multiple classification models; Multinomial Naive Bayes performed best, achieving 97.33% accuracy and 100% precision on the test set.
- Visualized class distribution and model performance with Matplotlib to guide feature selection and threshold tuning.
- Built an interactive Streamlit web app for real-time spam prediction from user-entered text.
- Deployed the project on GitHub with clear documentation for setup and reproducibility.

Text Summarizer *Minor Project*

- Built an **NLP** text-summarization pipeline using **HuggingFace Transformer** models to automatically condense long documents while preserving key information.
- **Scraped** and collected raw text data from web sources, then built a preprocessing pipeline to clean and prepare unstructured corpora for summarization.
- Applied tokenization techniques to structure input text for the transformer model, improving processing efficiency and output quality.
- Exposed the summarization pipeline as a REST API using **FastAPI**, enabling real-time summary generation from external requests.
- Tested the pipeline across varied document types, achieving consistent, high-quality summarization performance in deployment.

Intelligent AI Attendance System *Academic Project*

- Built a real-time **facial recognition** attendance system using **OpenCV** and deep learning, replacing manual logging.
- **Architected** a real-time facial recognition and attendance tracking REST API using **Flask**, OpenCV, and deep learning models to completely automate manual logging.
- Integrated OCR-based ID badge recognition to extract and verify user information, enabling an additional authentication layer.
- **Automated** software delivery and testing by implementing a robust **CI/CD** pipeline, ensuring seamless deployment updates and low-latency performance for the live video and audio processing feeds.
- Implemented real-time video processing pipelines for live face detection and identification with low-latency performance.
- Designed the data collection and model training framework used to enroll and recognize multiple users.

EDUCATION

B.Tech, Computer Science (AI & ML) *Expected 2027*

S R Institution of Management and Technology, Lucknow, UP

Intermediate (Class XII) *2023*

Lucknow Model Public Inter College, Lucknow, UP

CERTIFICATIONS

- Nexus Hackathon x DEVRhylme — Participant/Hackathon Certificate
- Java Programming — Certification
- Frontend Web Development — Certification